

## On the Roots of Quadratic Polynomials

The roots of a quadratic polynomial are determined entirely by its coefficients. For real coefficients the discriminant decides whether the roots are real or complex conjugates. The closed form is given by the well known quadratic formula shown below as a display equation.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

When the discriminant is negative the square root is imaginary and the two roots form a complex conjugate pair. This concludes the example.